

# Direct and Indirect Salt Effects on Homotypic Phase Separation

Matthew MacAinsh, Souvik Dey, Huan-Xiang Zhou 

Department of Chemistry, University of Illinois Chicago, Chicago, IL 60607, USA • Department of Physics, University of Illinois Chicago, Chicago, IL 60607, USA

 [https://en.wikipedia.org/wiki/Open\\_access](https://en.wikipedia.org/wiki/Open_access)
 Copyright information

 A newer version is available. [Read the latest version.](#)

Version of Record

November 12, 2024

Reviewed Preprint

v2 • October 21, 2024

Revised by authors

Reviewed Preprint

v1 • August 13, 2024

## eLife Assessment

In this potentially **important** study, the authors conducted atomistic simulations to probe the salt-dependent phase separation of the low-complexity domain of hnRNPA1 (A1-LCD). The authors have identified both direct and indirect mechanisms of salt modulation, provided explanations for four distinct classes of salt dependence, and proposed a model for predicting protein properties from amino acid composition. There is a range of opinions regarding the strength of evidence, with some considering the evidence as **incomplete** due to the limitations in the length and statistical errors of the computationally intense atomistic MD simulations.

<https://doi.org/10.7554/eLife.100282.2.sa4>

## Abstract

The low-complexity domain of hnRNPA1 (A1-LCD) phase separates in a salt-dependent manner. Unlike many intrinsically disordered proteins (IDPs) whose phase separation is suppressed by increasing salt concentrations, the phase separation of A1-LCD is promoted by > 100 mM NaCl. To investigate the atypical salt effect on A1-LCD phase separation, we carried out all-atom molecular dynamics simulations of systems comprising multiple A1-LCD chains at NaCl concentrations from 50 to 1000 mM NaCl. The ions occupy first-shell as well as more distant sites around the IDP chains, with Arg sidechains and backbone carbonyls the favored partners of  $\text{Cl}^-$  and  $\text{Na}^+$ , respectively. They play two direct roles in driving A1-LCD condensation. The first is to neutralize the high net charge of the protein (+9) by an excess of bound  $\text{Cl}^-$  over  $\text{Na}^+$ ; the second is to bridge between A1-LCD chains, thereby fortifying the intermolecular interaction networks in the dense phase. At high concentrations, NaCl also indirectly strengthens  $\pi$ - $\pi$ , cation- $\pi$ , and amino- $\pi$  interactions, by drawing water away from the interaction partners. Therefore, at low salt, A1-LCD is prevented from phase separation by net charge repulsion; at intermediate concentrations, NaCl neutralizes enough of the net charge while also bridging IDP chains to drive phase separation. This drive becomes even stronger at high salt due to strengthened  $\pi$ -type interactions. Based on this understanding, four classes of salt dependence of IDP phase separation can be predicted from amino-acid composition.

## Introduction

Biomolecular condensates formed via liquid-liquid phase separation (LLPS) mediate a variety of cellular functions such as biogenesis of the ribosome and stress response.<sup>1,2</sup> The driver for phase separation or condensation is intermolecular interactions, including electrostatic, hydrogen bonding,  $\pi$ - $\pi$ , cation- $\pi$ , amino- $\pi$ , and hydrophobic.<sup>3–5</sup> Salt can tune all these interactions and thus exert significant effects on phase separation. While the screening effect of salt on electrostatic interactions is well-known, its effects on other types of interactions may be indirect and perhaps are less appreciated. In particular, high salt strengthens hydrophobic interactions by increasing the surface tension of water.<sup>6,7</sup> Because salt can exert disparate effects on different types of interactions, it can be used as a perturbation to dissect the relative importance of these interactions in phase separation.<sup>8,9</sup> Inside cells, proteins can encounter varying salt conditions at different locations or at different times, and therefore the drive for their phase separation can span a wide range.

Numerous studies of salt effects on protein phase separation have been reported. The most typical effect is the suppression of phase separation by screening electrostatic attraction, both for homotypic systems<sup>10–18</sup> and for heterotypic systems.<sup>8,19</sup> However, salt can also promote phase separation, although the mechanism is not always clear.<sup>17,20–29</sup> In particular, Krainer et al.<sup>28</sup> observed a “reentrant” salt effect on the phase separation of five intrinsically disordered proteins (IDPs): phase separation occurs without salt, is prohibited by medium salt, and reemerges at high salt. They attributed the reemergence of phase separation to strengthened  $\pi$ -type and hydrophobic interactions that overcompensate weakened electrostatic attraction. A unifying understanding of how salt affects the phase separation of IDPs is still lacking. For example, it is an open question whether salt effects on phase separation can be predicted from the protein sequence. Deep knowledge, in particular at the atomic level, of how salt affects intermolecular interactions and, ultimately, phase separation is required.

The low-complexity domain of hnRNPA1 (A1-LCD) represents another IDP where an atypical salt effect was reported.<sup>17</sup> The full-length protein, comprising folded domains along with the LCD, phase separates in the absence of salt and the tendency to phase separate is reduced upon adding salt, thus exhibiting the typical salt effect. A screening mechanism, specifically of electrostatic attraction between the LCD and the folded domains, is supported by small-angle X-ray scattering and coarse-grained molecular dynamics (MD) simulations. In contrast, A1-LCD does not phase separate without salt and starts to do so only after 100 mM NaCl is added. An earlier study revealed that  $\pi$ -types of interactions, mediated by aromatic residues, drive the phase separation of A1-LCD.<sup>30</sup> A follow-up study, based on charge mutations, further showed that the net charge plays a strong suppressive role.<sup>31</sup> The saturation concentration ( $C_{\text{sat}}$ ) for phase separation is minimum at a net charge near 0, and increases by two orders of magnitude, signifying an enormous weakening of the drive for phase separation, when the net charge moves away from neutrality in either direction. In a more recent study, salt promoted the homotypic phase separation of both A1-LCD and FUS-LCD, but suppressed the heterotypic phase separation of their mixture.<sup>8</sup> The salt effect on the phase separation of A1-LCD was modeled by a Debye-Hückel potential in coarse-grained simulations.<sup>32</sup> Similarly, the salt effect on the phase separation of another IDP, Ddx4, was analyzed using the random phase approximation based on a coarse-grained representation.<sup>18,33,34</sup> Coarse-grained simulations with explicit water and ions have been used to study salt effects in both homotypic and heterotypic phase separation.<sup>35</sup>

All-atom MD simulations can uniquely provide mechanistic insight into the driving force and properties of biomolecular condensates.<sup>5</sup> For example, these simulations showed that ATP, a small molecule with a -4 charge, bridges between positively charged IDP chains in driving phase separation.<sup>36</sup> The intermolecular interactions quickly break and reform, explaining why the

condensates can rapidly fuse despite very high macroscopic viscosity. Similarly, quick breakup and reformation of salt bridges in a heterotypic condensate allow the protein molecules to be extremely dynamic in a highly viscous environment.<sup>19</sup> Recently, all-atom MD simulations provided explanations for wide variations in phase equilibrium and material properties among condensates of tetrapeptides with different amino-acid compositions.<sup>37</sup> These and other simulations<sup>38</sup> show that all attractive residue-residue contacts contribute to the drive for phase separation.

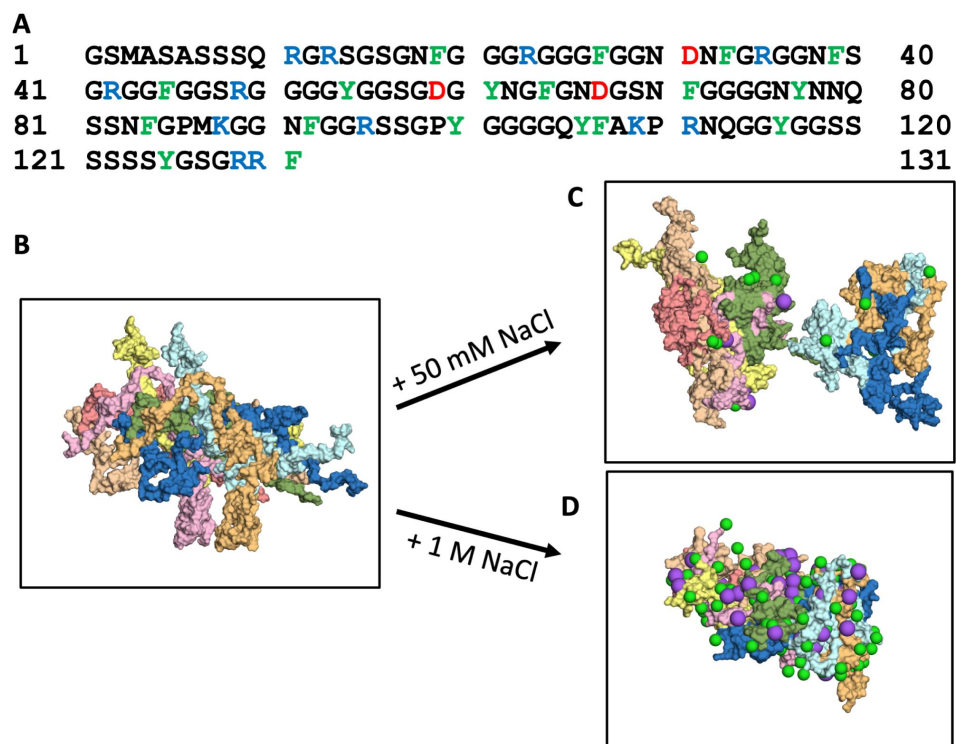
Here we study salt effects on A1-LCD condensation by all-atom MD simulations. The simulations reveal two direct effects and one indirect effect of NaCl: neutralization of net charge and bridging between protein chains as well as strengthening of  $\pi$ -type interactions by drawing water away from the interaction partners. We also present a unified picture of salt dependences of phase separation by defining four distinct classes and predict these classes from amino-acid composition.

## Results

### Salt condenses A1-LCD and increases inter-chain interactions

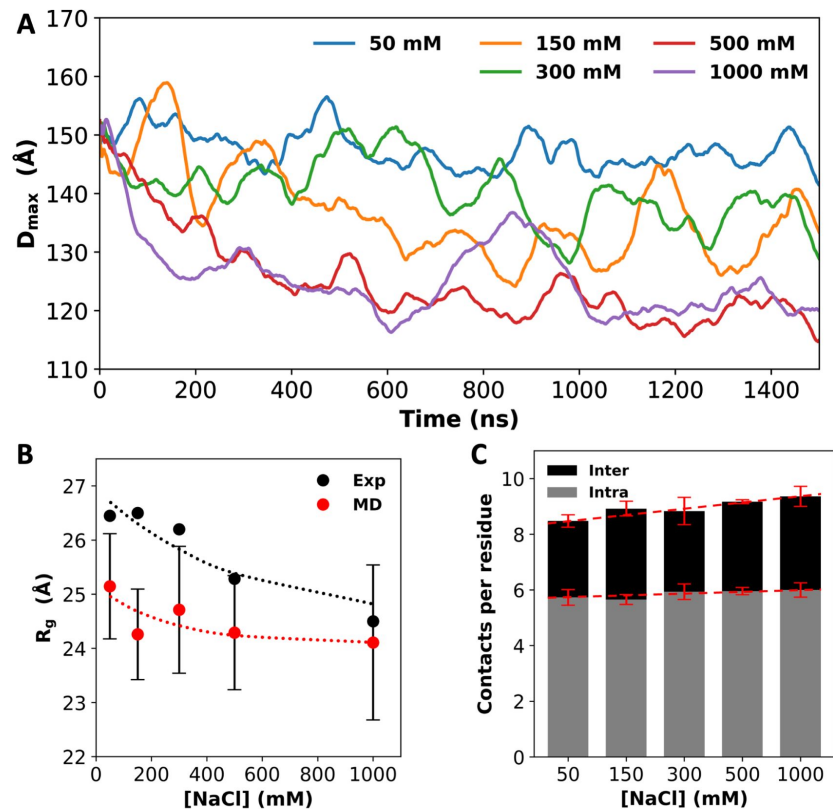
The 131-residue A1-LCD is comprised mostly of Gly and Ser (51 and 22, respectively), followed by 18 aromatic residues (11 Phe and 7 Tyr), 17 residues with sidechain amides (13 Asn and 4 Gln), and 15 charged residues (10 Arg, 2 Lys, and 3 Asp), with a large net charge of +9 (**Figure 1A**). Using initial conformations of A1-LCD from the previous single-copy simulations,<sup>39</sup> we built an 8-copy model for the dense phase (with an initial concentration of 3.5 mM) at five NaCl concentrations ranging from 50 to 1000 mM (**Figure 1B**). This initial concentration is close to the measured concentration at the critical point,<sup>30</sup> thereby facilitating the comparison between low salt (where phase separation does not occur) and high salt (where phase separation does occur). Each system was simulated in four replicates for 1.5  $\mu$ s each. All the results reported here are averages over the four replicate simulations. From the same initial configuration with very few inter-chain contacts, the systems maintain the initial loose configuration at 50 mM NaCl (**Figure 1C**) but condense noticeably at 1000 mM NaCl (**Figure 1D**). At low salt, the protein chains have a tendency to fill the simulation box, occasionally spanning the entire box in one or two of the three orthogonal directions and exhibiting larger-scale reconfiguration based on root-mean-square-fluctuation (RMSF) calculation (Supplementary Figure 1). In contrast, at high salt, the chains aggregate into an ellipsoidal particle, with the RMSF reduced by 27%.

We also quantified the contrast between low and high salt by calculating  $D_{\max}$ , the maximum side length of the rectangular box that parallels the simulation box and circumscribes the multi-chain system. The  $D_{\max}$  values are close to the initial value ( $\sim 150$  Å) at low salt (50 mM NaCl), but decrease to  $\sim 135$  Å at intermediate salt (150 and 300 mM NaCl) and further to  $\sim 120$  Å at high salt (500 and 1000 mM NaCl; **Figure 2A**). Overall, the systems show successive condensation on going from low salt to intermediate salt to high salt. Modeling the condensed particle as an ellipsoid with the principal diameters given by the maximum dimensions in the three orthogonal directions, we can estimate its concentration to be 23 mM, which is similar to the measured concentration in the dense phase.<sup>30</sup> A small part of the reason for the growing condensation is the compaction of the individual chains. The average radius of gyration ( $R_g$ ) shows a decreasing trend with increasing salt (**Figure 2B**), which matches the experimental data.<sup>17</sup> Correspondingly, the average number of intrachain contacts per residue increases slightly, by 5%, when the salt concentration is increased from 50 mM to 1000 mM (**Figure 2C**). However, the main driver of the condensation is inter-chain interactions, with inter-chain contacts per residue increasing by 23% over the same salt range. Clearly, high salt induces A1-LCD condensation, with an increased number of interactions between protein chains. Below we present the molecular mechanisms for these effects.



**Figure 1.**

Amino-acid sequence of A1-LCD and molecular dynamics simulations of its condensation. (A) Amino-acid sequence. (B) First frame and (C, D) a frame at 1000 ns from 1.5- $\mu$ s simulations of the 8-chain systems at low and high salt. In all figures,  $\text{Cl}^-$  or  $\text{Na}^+$  ions are represented by green and magenta spheres, respectively.



**Figure 2.**

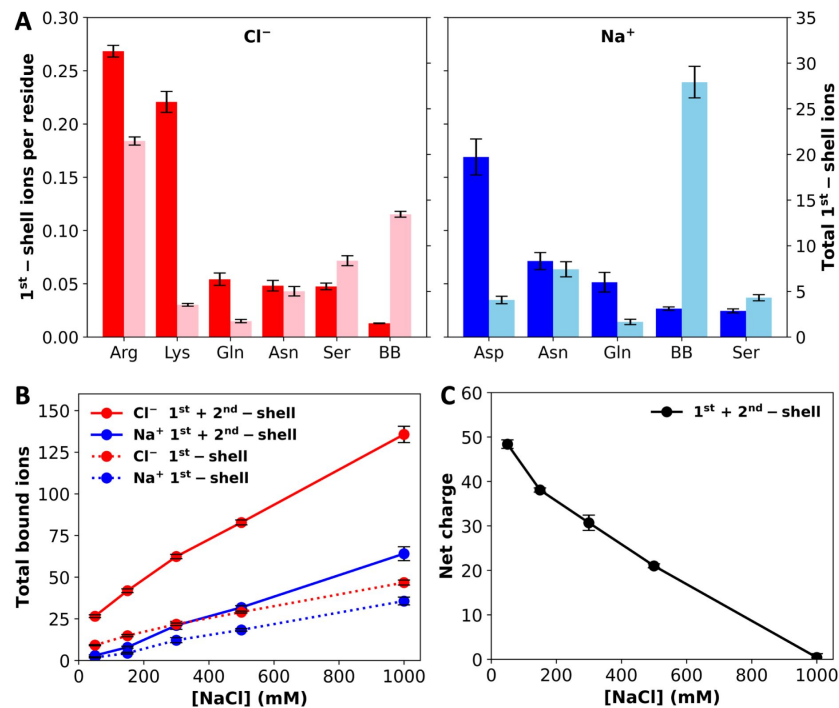
Salt effects on A1-LCD condensation and inter-chain interactions. (A)  $D_{\max}$  values, averaged over four replicates, as a function of simulation time. (B) Radii of gyration ( $R_g$ ) from small-angle X-ray scattering<sup>17</sup> and from MD simulations. Dotted curves are drawn to guide the eye. (C) Average number of inter- or intra-chain contacts per residue at each salt concentration. Dashed lines are drawn to show trends. Error bars represent standard deviations among four replicate simulations.

## Ions selectively bind to backbone and sidechain sites to neutralize the net charge of A1-LCD

The large net charge of A1-LCD implies significant electrostatic repulsion between chains. Potentially salt ions can neutralize the net charge. To investigate ion-protein binding, we calculated radial distribution functions (RDFs) of  $\text{Cl}^-$  and  $\text{Na}^+$  around polar groups of A1-LCD. At 1000 mM NaCl, the RDFs of  $\text{Cl}^-$  show a strong 1<sup>st</sup> peak at 3.2 Å around Arg and Lys sidechain nitrogens and a moderate 1<sup>st</sup> peak around the Gln and Asn sidechain nitrogens and the Ser sidechain oxygen (Supplementary Figure 2A). Each Arg sidechain often coordinates two  $\text{Cl}^-$  ions simultaneously, but each Lys sidechain coordinates only one  $\text{Cl}^-$  ion. A 2<sup>nd</sup> peak, at ~5 Å, is also observed around Arg and Lys sidechain nitrogens. Of the remaining polar groups, only the  $\text{Cl}^-$  RDFs around the Tyr sidechain oxygen and the backbone nitrogen show a weak 1<sup>st</sup> peak (Supplementary Figure 2B). We used cutoffs of 4 and 6.4 Å, respectively, to define 1<sup>st</sup>-shell and 2<sup>nd</sup>-shell  $\text{Cl}^-$  binding. On a per-residue basis, Arg and Lys sidechains coordinate the most 1<sup>st</sup>-shell  $\text{Cl}^-$  ions, reaching ~0.25 ions per residue (Figure 3A, left panel). In comparison, each Gln, Asn, or Ser residue, on average, coordinates ~0.05  $\text{Cl}^-$  ions. Given the large numbers of Arg and Ser residues in the A1-LCD sequence, these two residue types coordinate the most 1<sup>st</sup>-shell  $\text{Cl}^-$  ions, 21.5 and 8.4, respectively, in the 8-chain system at 1000 mM NaCl. Although the RDF around backbone nitrogens has only a weak 1<sup>st</sup> peak, given their large number (131 per chain), they actually coordinate 13.4, or the second most 1<sup>st</sup>-shell  $\text{Cl}^-$  ions. Among these 13.4  $\text{Cl}^-$  ions, 55% are coordinated with Gly residues. A large part of this high percentage is due to the enrichment of Gly (39% of all residues) in the A1-LCD sequence, but Gly is additionally favored for its lack of sidechain, which allows the close approach of ions to the backbone.

$\text{Na}^+$  shows a very strong preference for the Asp carboxyl in 1<sup>st</sup>-shell coordination, with a peak RDF value of 9.2 at 2.2 Å (Supplementary Figure 3A). Each Asp sidechain carboxyl typically coordinates a single  $\text{Na}^+$  ion, usually in a bifurcated geometry. In addition,  $\text{Na}^+$  shows a strong preference in 1<sup>st</sup>-shell coordination with Asn and Gln sidechain amide oxygens and the backbone carbonyl oxygen as well as a moderate preference with the Ser sidechain oxygen. A 2<sup>nd</sup> peak, at ~4.2 Å, is also observed in the  $\text{Na}^+$  RDF around the Asp carboxyl. Of the remaining polar groups, only a weak 1<sup>st</sup> peak is seen in the RDF around the Tyr sidechain oxygen (Supplementary Figure 3B). We used cutoffs of 3 and 5.4 Å, respectively, to define 1<sup>st</sup>-shell and 2<sup>nd</sup>-shell  $\text{Na}^+$  binding. On a per-residue basis, the Asp sidechain coordinates the most 1<sup>st</sup>-shell  $\text{Na}^+$  ions, reaching 0.17 ions per residue (Figure 3A, right panel). The per-residue  $\text{Na}^+$  ion number reduces to ~0.06 for Asn and Gln sidechain oxygens and further to ~0.03 for the backbone oxygen and Ser sidechain oxygen. Again, due to their large number, backbone oxygens coordinate a very large number, 27.9, of 1<sup>st</sup>-shell  $\text{Na}^+$  in the 8-chain system at 1000 mM NaCl. This number dwarfs the counterparts for sidechains, the largest of which are 7.4, 4.1, and 4.3 respectively, for Asn, Asp, and Ser. For coordination with the backbone, similar to the case with  $\text{Cl}^-$ , 55% of the 27.9  $\text{Na}^+$  ions involve Gly residues, again reflecting the fact that this amino acid allows the close approach of ions to the backbone. On the other hand, the largest number of 1<sup>st</sup>-shell  $\text{Cl}^-$  ions are coordinated with Arg sidechains but the largest number of 1<sup>st</sup>-shell  $\text{Na}^+$  ions are coordinated with backbone carbonyls. Whereas each Arg sidechain often coordinates two  $\text{Cl}^-$  ions, multiple backbone carbonyls often coordinate a single  $\text{Na}^+$  ion, with at least two backbone carbonyl partners for 30.6 of the 64.1 bound  $\text{Na}^+$  ions (1<sup>st</sup>- and 2<sup>nd</sup>-shell).

In Figure 3B, we display the total numbers of 1<sup>st</sup>-shell  $\text{Cl}^-$  and  $\text{Na}^+$  ions in the 8-chain systems as a function of NaCl concentration. Both the 1<sup>st</sup>-shell  $\text{Cl}^-$  and  $\text{Na}^+$  ions increase with increasing NaCl concentration, and  $\text{Cl}^-$  ions always outnumber  $\text{Na}^+$  ions. The difference remains almost constant, with around 10 more 1<sup>st</sup>-shell  $\text{Cl}^-$  ions than 1<sup>st</sup>-shell  $\text{Na}^+$  ions. This difference is not enough to neutralize the total charge, +72, on the protein chains. However, when 2<sup>nd</sup>-shell ions are also included, the excess of bound  $\text{Cl}^-$  ions over bound  $\text{Na}^+$  ions grows quickly with increasing NaCl



**Figure 3.**

Levels of ion binding at different NaCl concentrations. (A) Number of 1<sup>st</sup>-shell ions, with (darker colors; tick marks on the left vertical axis) and without (lighter colors; tick marks on the right vertical axis) normalization by the number of residues of a given amino-acid type, at 1000 mM NaCl. (B) Total number of 1<sup>st</sup>-shell only or 1<sup>st</sup>- and 2<sup>nd</sup>-shell ions. (C) Net charge of the system with 1<sup>st</sup>- and 2<sup>nd</sup>-shell ions included. Error bars represent standard deviations among four replicate simulations.

concentration. Correspondingly, the net charge of the system reduces to zero at 1000 mM NaCl (**Figure 3C**). Net charge repulsion explains why the chains maintain their initial loose configuration in the simulations at 50 mM NaCl (**Figure 1C**) and the observed absence of phase separation at this low salt concentration.<sup>17</sup> At high salt, the protein net charge is completely neutralized by 1<sup>st</sup>- and 2<sup>nd</sup>-shell ions; hence the protein chains condense (**Figure 1D**) and phase separation was readily observed.<sup>17</sup>

## Ions act as bridges between protein chains to drive condensation

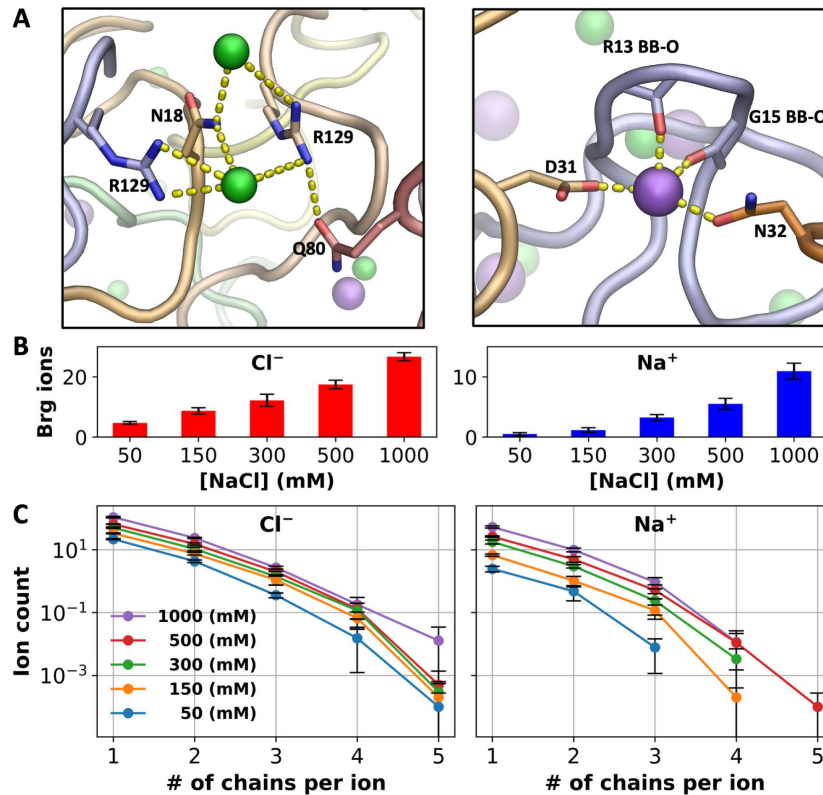
In addition to charge neutralization, we suspected that ions could also fortify intermolecular interactions by bridging between proteins, similar to the role played by ATP molecules in driving phase separation of positively charged IDPs.<sup>36</sup> Indeed, we found that Cl<sup>-</sup> has a tendency to bind with Arg and other sidechains from multiple chains and likewise, Na<sup>+</sup> has a tendency to bind with backbone carbonyls and sidechain oxygens from multiple chains (**Figure 4A**).

To quantify this tendency, we calculated the number of bridging Cl<sup>-</sup> or Na<sup>+</sup> ions, i.e., those that bind (both 1<sup>st</sup>- and 2<sup>nd</sup>-shell) to residues on more than one A1-LCD chain (**Figure 4B**). Close to 20% of all 1<sup>st</sup>- and 2<sup>nd</sup>-shell ions bridge between A1-LCD chains. For both Cl<sup>-</sup> and Na<sup>+</sup>, the average number of bridging ions increases with increasing salt, but the pace of increase is much greater for Na<sup>+</sup> than for Cl<sup>-</sup>, commensurate with the trend shown by the total number of bound ions of each type (**Figure 3B**). The number of bridging ions is 4.7 for Cl<sup>-</sup> and only 0.5 for Na<sup>+</sup> in the 8-chain system at 50 mM NaCl, and increases to 26.6 and 10.9, respectively, at 1000 mM NaCl. The greater fold change of bridging Na<sup>+</sup> ions (23-fold, vs 6-fold for Cl<sup>-</sup>) is also apparent when we break the bridging ions according to the number of A1-LCD chains being bridged (**Figure 4C**). For Cl<sup>-</sup>, the curves plotting the ion count against the number of bridged chains are close to each other among the different NaCl concentrations, but the counterparts for Na<sup>+</sup> are more spread out.

These disparate effects of salt concentration on chain bridging by Cl<sup>-</sup> and Na<sup>+</sup> can be explained by the difference in coordination properties between the two ion types presented above. Cl<sup>-</sup> strongly prefers Arg sidechains and, even at low salt, occupies a large number of sites lined by them, of which ~20% are bridging sites. However, there is a relatively limited supply of Arg sidechains (a total of 80 in the 8-chain system) and hence the fold change in bridging Cl<sup>-</sup> ions upon salt increase is somewhat tempered. In contrast, although Na<sup>+</sup> prefers Asp sidechains, there are very few of those in the system; instead, Na<sup>+</sup> predominantly binds to backbone carbonyls. As the latter binding is relatively weak, the number of bridging Na<sup>+</sup> ions is very small at low salt. However, since there is a large supply of backbone carbonyls (a total of 1048 in the 8-chain system), at high salt, Na<sup>+</sup> ions bind to a portion of these backbone carbonyls and bridge A1-LCD chains. Of the total 10.9 Na<sup>+</sup> bridging ions, 6.4 do so through backbone carbonyls of at least two chains. Consequently, at low salt, chain bridging is dominated by Cl<sup>-</sup> (Na<sup>+</sup> only 9% of bridging ions), but at high salt, Na<sup>+</sup> becomes more even (30% of bridging ions) with Cl<sup>-</sup> in bridging A1-LCD chains. An important reason for the latter is Na<sup>+</sup> coordination by backbone carbonyls, especially those of Gly residues (**Figure 4A**).

## Salt also contributes to condensation indirectly by strengthening $\pi$ -type interactions

As noted above, the number of inter-chain interactions increases by 23% when the NaCl concentration increases from 50 mM to 1000 mM. While this result could be accounted for by the two direct salt effects presented so far, i.e., charge neutralization and chain bridging, which act to condense A1-LCD, there are additional factors. A breakdown of inter-chain sidechain interactions into different types reveals that when the salt concentration is increased from 50 to 1000 mM, the number of salt bridges per chain remains nearly constant, while the numbers of cation- $\pi$ ,  $\pi$ - $\pi$ , and amino- $\pi$  interactions increase by 17%, 26%, and 39%, respectively (**Figure 5A**). That is, as A1-LCD is condensed at high salt, there is an overall increase in inter-chain interactions, but this



**Figure 4.**

Bridging ions. (A) Examples of chain bridging by Cl<sup>-</sup> and Na<sup>+</sup>; Cl<sup>-</sup> ions are coordinated to Arg and other sidechains whereas Na<sup>+</sup> ions are coordinated to both backbone carbonyls (including from Gly) and sidechain oxygens. (B) Average number of Cl<sup>-</sup> or Na<sup>+</sup> ions engaged in bridging between A1-LCD chains. (C) Average number of ions bound in 1<sup>st</sup>- and 2<sup>nd</sup>-shell sites lined by a given number of A1-LCD chains. Error bars represent standard deviations among four replicate simulations.

increase is limited to  $\pi$ -types of interactions and excludes salt bridges. At increasing salt, more  $\pi$ -types of interactions are formed while no new salt bridges are formed, suggesting a strengthening of the former interactions.

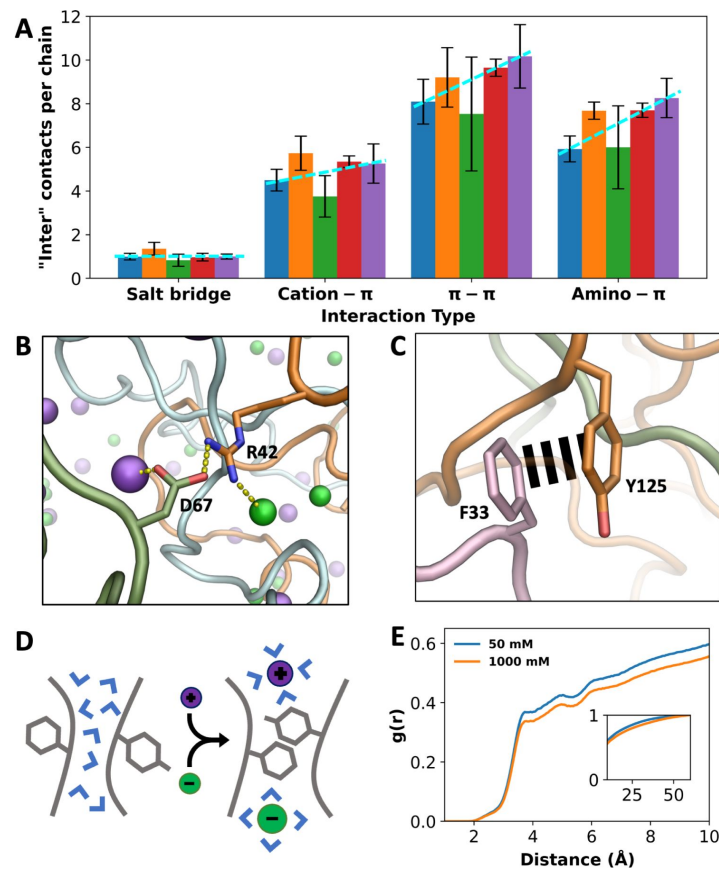
The apparent null effect of salt on salt bridge formation can be attributed to the canceling of two opposing effects: chain condensation by the direct effects of salt may potentially shorten the distances between salt-bridge partners and thereby strengthen salt bridges but competition of ions with the salt-bridge partners for coordination (Supplementary Figures 2A and 3A) may potentially weaken salt bridges. Indeed, at 1000 mM NaCl, cationic and anionic partners in a salt bridge are often found to also coordinate  $\text{Cl}^-$  and  $\text{Na}^+$ , respectively (Figure 5B). In comparison, one partner in cation- $\pi$  interactions and both partners in  $\pi$ - $\pi$  and amino- $\pi$  interactions have limited abilities to coordinate with ions, so the effects from ion competition are tempered.

For  $\pi$ -types of interactions, instead of a weakening mechanism through ion competition, salt may exert a strengthening effect by drawing water away from the interaction partners (Figure 5D). We calculated the RDFs of water around Tyr sidechains that form  $\pi$ -types of interactions and found decreases in water density when the salt concentration is increased from 50 to 1000 mM (Figure 5E and Supplementary Figure 4A-C). The decreases in water density follow the order cation- $\pi$  <  $\pi$ - $\pi$  < amino- $\pi$ , which is the same order as the corresponding increases in these three types of interactions. As a control, no decrease in water density is seen around Asp sidechains that form salt bridges (Supplementary Figure 4D). Water can interfere with and weaken  $\pi$ -types of interactions; by drawing water away from the interaction partners, high salt strengthens these interactions and thereby indirectly contributes to condensation. Recently the withdrawal of water from  $\pi$ -types of interaction partners in FUS-LCD condensates has been directly demonstrated by Raman spectroscopy.<sup>40</sup>

## Discussion

We have shown in atomistic detail the actions of ions in the condensation of A1-LCD over a wide range of NaCl concentrations. The MD simulations reveal that NaCl has both direct and indirect effects in driving phase separation. The first direct effect is to neutralize the net charge of A1-LCD and thereby attenuate net charge repulsion. The second direct effect is to bridge between protein chains and thereby fortify intermolecular interaction networks. In addition, high salt strengthens  $\pi$ -types of interactions by drawing water away from the interaction partners, thereby also indirectly driving phase separation. The net result is that, while phase separation of A1-LCD is prevented by net charge repulsion at low salt, it is enabled at intermediate salt through charge neutralization and chain bridging by ions. The drive for phase separation becomes even stronger at high salt, where  $\pi$ -types of interactions are strengthened.

These findings broaden our understanding of the roles of charges and ions in phase separation. That high net charge is a suppressive factor is highlighted by the strong effect of A1-LCD charge mutations on  $C_{\text{sat}}$ .<sup>31</sup> Here we have shown that this suppressive action can be countered by the addition of salt, which exerts both direct and indirect effects. One of these direct effects, i.e., neutralization of net charge, can be treated by a Debye-Hückel potential in coarse-grained simulations.<sup>32</sup> However, our all-atom explicit-solvent simulations have revealed not only an additional direct effect, i.e., bridging between protein chains, but also an indirect effect, i.e., strengthening of  $\pi$ -type interactions. In lattice Monte Carlo simulations of a system of charged homopolymer chains plus counterions with Coulomb interactions between all charges, counterions were found to occupy sites between chains and produce effective chain-chain attraction;<sup>41</sup> this effect might be viewed as a primitive form of bridging. Krainer et al.<sup>28</sup> observed a reentrant salt effect on the phase separation of IDPs, and attributed the reemergence of phase separation at high salt to strengthened  $\pi$ -type and hydrophobic interactions that overcompensate weakened electrostatic attraction. Their conclusion was based on the salt



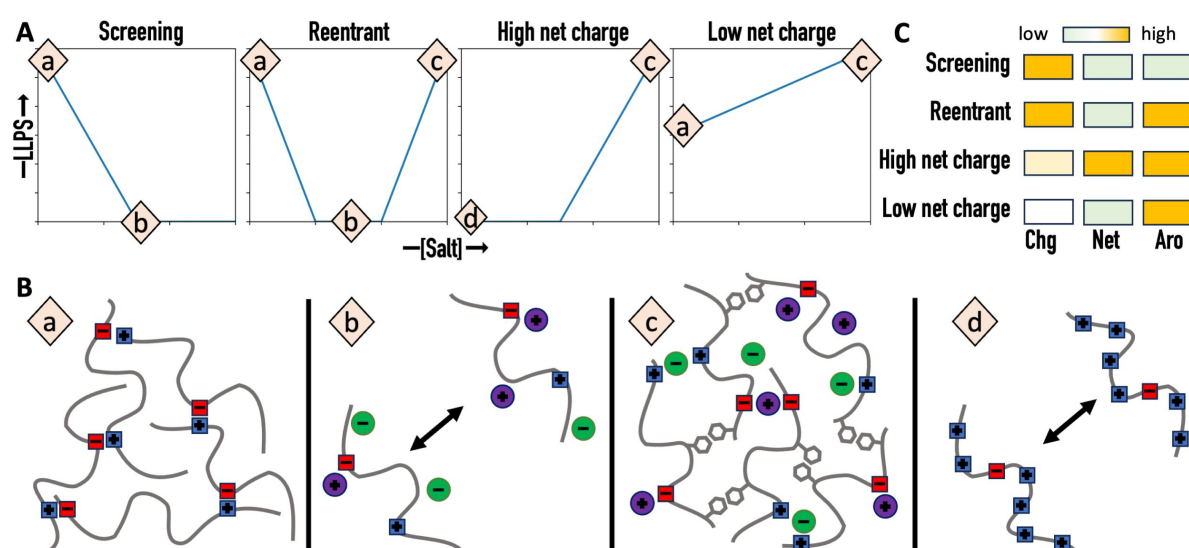
**Figure 5.**

Indirect effects of ions on different types of interactions. (A) Number of inter-chain contacts per chain for each interaction type. Bars from left to right correspond to increasing salt concentrations (50, 150, 300, 500, and 1000 mM). Dashed lines are drawn to show trends. Error bars represent standard deviations among four replicate simulations. (B) An inter-chain salt bridge, with ion coordination by the partner sidechains. (C) An inter-chain  $\pi$ - $\pi$  interaction, free of ion coordination. (D) Schematic showing a  $\pi$ - $\pi$  interaction facilitated by high salt, via drawing water away from the interaction partners. (E) Radial distribution functions of water around Tyr residues that interact with Phe, Arg, Lys, Gln, and Asn. Lower values at high salt demonstrate water withdrawal. Inset shows radial distribution functions approaching 1.

dependences of the potentials of mean force calculated for pairs of sidechains of various types. Instead of a single pair of sidechains, our simulations are on multiple copies of protein chains. We show that, at high salt, the number of salt bridges remains constant while the numbers of  $\pi$ -types of interactions are elevated in the multi-chain system. Our simulations further reveal that high salt achieves the latter effect by drawing water away from  $\pi$ -interaction partners.

The present work puts us in a position to paint a unified picture of salt dependences of homotypic phase separation (**Figure 6A**, B and **Table 1**). The salt dependence that is most often reported is screening, where salts weaken the electrostatic attraction between protein chains and thereby suppress phase separation.<sup>10–18</sup> The reentrant salt dependence reported by Krainer et al.<sup>28</sup> can be seen as an extension of the screening scenario, whereby high salt overcompensates the screening effect by strengthening  $\pi$ -type and hydrophobic interactions and leads to reemergence of phase separation. The high net charge scenario represented by A1-LCD is similar to the reentrant scenario at high salt but differs from it at low salt, where phase separation is prevented by net charge repulsion.<sup>17,20,22–27,29</sup> In this class of salt dependence, a certain amount of salt is required to start phase separation. For A1-LCD, this minimum salt is  $\sim 100$  mM NaCl.<sup>17</sup> The recombinant mussel foot protein-1 (RMFP-1) presents an extreme example, which has a net charge of +24 over a sequence length of 121 and requires up to 700 mM NaCl to begin phase separation.<sup>22</sup> The fourth class of salt dependence is a variation of the preceding one; here the net charge is low and no significant net charge repulsion is expected, so the protein can phase separate even without salt. For example, FUS-LCD has a low net charge of -2 and phase separates without salt.<sup>21</sup> The difference between the high and low net charge scenarios is well captured by the phase-separation behaviors of TDP43-LCD at two pHs.<sup>24</sup> At pH 7, this protein has a relatively low net charge of +5 and phase separates without salt; the salt dependence thus belongs to the low net charge class. When pH is lowered to 4, the six His residues in the purification tag become positively charged, raising the net charge to +11; now phase separation requires 300 mM NaCl and the salt dependence switches to the high net charge class.

We can not only rationalize the four distinct classes of salt dependence but also start to predict them from the amino-acid composition of the protein (**Figure 6C** and **Table 1**). The foregoing physical understanding suggests three determinants of salt dependence class: (1) the total number of charged residues, which determines the contribution of electrostatic interactions to the drive for phase separation and the importance of salt screening; (2) the net charge, which determines the magnitude of net charge repulsion; and (3) the total number of aromatic residues, which determines whether phase separation reemerges at high salt. We predict that the screening class occurs when the charged content is high but the net charge and aromatic content are low (**Figure 6C**). The other three classes of salt dependence all require a high aromatic content. For the reentrant class, the increase in aromatic content is the only difference from the screening class. For the high net charge class, the net charge obviously has to be high, but that could also mean at least a moderately high charged content. Finally the low net charge class is predicted when the net charge is low but no bias is required of the charged content. The amino-acid compositional data in **Table 1** validate these predictions. For example, of the 17 proteins in the classes of salt dependence that call for a high aromatic content, 12 (or 71%) actually have this feature (aromatic content > an 8% threshold). In contrast, of the 10 proteins in the class of salt dependence (i.e., “screening”) that calls for a low aromatic content, only 2 (or 20%) have an aromatic content above the threshold. In the latter two cases it remains to be tested whether phase separation would reemerge at much higher salt (and thereby resulting in a reclassification to reentrant). Likewise, of the 10 proteins in the class of salt dependence (i.e., “high net charge”) that calls for a high net charge, 7 (or 70%) actually have this feature (net charge > a 6% threshold). In contrast, of the 17 proteins in the classes of salt dependence that call for a low net charge, none has a net charge above the threshold. Lastly, of the 15 proteins in the classes of salt dependence that call for a high charge content, 11 (or 73%) actually have this feature (charged content > a 20% threshold). In comparison, of the 12 proteins in the classes of salt dependence that do not require a high charge content, only 2 (or 17%) have a charge content above the threshold.



**Figure 6.**

Four classes of salt dependence and their prediction from amino-acid composition. (A) Salt dependences of liquid-liquid phase separation (LLPS). (B) Charge-charge and  $\pi$ -type interactions and their regulation by salt. (a) Significant Charge-charge attraction. (b) Screening of charge-charge attraction by salt. (c) Strengthening of  $\pi$ -type interactions by high salt. (d) Repulsion due to high net charge. (C) Distinctions of the four classes of salt dependence by three determinants: charged content ("Chg"), net charge ("Net"), and aromatic content ("Aro").

| Protein                 | Length | Charges (+/-/net) <sup>a</sup>                | Aromatic <sup>b</sup> | Salt (mM)     | Ref |
|-------------------------|--------|-----------------------------------------------|-----------------------|---------------|-----|
| Screening               |        |                                               |                       |               |     |
| FIB-1                   | 352    | <b>55<sup>c</sup></b> / <b>34</b> / 21        | 24                    | NaCl 50-250   | 10  |
| PolyQ                   | 729    | 38 / 34 / 4                                   | 35                    | NaCl 0-150    | 11  |
| Ddx4                    | 236    | <b>32</b> / <b>36</b> / -4                    | <b>22</b>             | NaCl 100-500  | 12  |
| HP1 $\alpha$            | 206    | <b>33</b> / <b>41</b> / -8                    | 14                    | NaCl 25-150   | 13  |
| LAF-1                   | 708    | <b>86</b> / <b>88</b> / -2                    | 55                    | NaCl 125-400  | 14  |
| LAF-1 RGG               | 191    | <b>28</b> / <b>22</b> / 6                     | 14                    | NaCl 125-300  | 14  |
| Oleo30G                 | 139    | 15 / 12 / 3                                   | 7                     | NaCl 35-280   | 15  |
| FMRP-LCD                | 188    | <b>37</b> / <b>28<sup>e</sup></b> / 9         | 6                     | NaCl 0-150    | 16  |
| hnRNPA1                 | 314    | <b>42</b> / <b>34</b> / 8                     | <b>33</b>             | NaCl 50-300   | 17  |
| pY-Caprin1 <sup>d</sup> | 103    | <b>16</b> / <b>17</b> / -1                    | 4                     | NaCl 100-500  | 18  |
| Reentrant               |        |                                               |                       |               |     |
| FUS                     | 526    | 51 / 37 / 14                                  | <b>52</b>             | KCl 50-2700   | 28  |
| TDP-43                  | 414    | <b>40</b> / <b>44</b> / -4                    | <b>36</b>             | KCl 50-2700   | 28  |
| Brd4                    | 1362   | <b>175</b> / <b>150</b> / 25                  | 48                    | KCl 50-2150   | 28  |
| Sox2                    | 317    | 34 / 21 / 13                                  | 19                    | KCl 50-2150   | 28  |
| A11                     | 505    | <b>51</b> / <b>50</b> / 1                     | <b>41</b>             | NaCl 22.5-500 | 28  |
| High net change         |        |                                               |                       |               |     |
| A1-LCD                  | 131    | 12 / 3 / <b>9</b>                             | <b>18</b>             | NaCl 50-300   | 17  |
| Lysozyme <sup>e</sup>   | 129    | <b>17</b> / <b>9</b> / <b>8</b>               | <b>12</b>             | NaCl 514-1198 | 20  |
| RMFP-1                  | 121    | 24 / 0 / <b>24</b>                            | <b>24</b>             | NaCl 100-500  | 22  |
| UBQLN2                  | 624    | 31 / 40 / -9                                  | 22                    | NaCl 100-300  | 23  |
| UBQLN2 (450-624)        | 175    | 4 / 8 / -4                                    | 5                     | NaCl 50-200   | 23  |
| TDP43-LCD (pH 4)        | 148    | 14 <sup>f</sup> / 3 / <b>11</b>               | <b>12</b>             | NaCl 0-300    | 24  |
| HBP-2 (pH 5.5)          | 193    | <b>30<sup>f</sup></b> / <b>13</b> / <b>17</b> | <b>20</b>             | NaCl 50-500   | 25  |
| Caprin1                 | 103    | 16 / 3 / <b>13</b>                            | <b>11</b>             | NaCl 0-2000   | 26  |
| Prp-LCD                 | 122    | 11 / 1 / <b>10</b>                            | <b>11</b>             | NaCl 150-750  | 27  |
| RLP3 <sub>8</sub>       | 120    | 12 / 8 / 4                                    | 8                     | NaCl 0-2000   | 29  |
| Low net charge          |        |                                               |                       |               |     |
| FUS-LCD                 | 136    | 0 / 2 / -2                                    | <b>20</b>             | NaCl 0-250    | 21  |
| TDP43-LCD (pH 7)        | 148    | 8 / 3 / 5                                     | <b>12</b>             | NaCl 0-300    | 24  |

<sup>a</sup>Charges are listed as the number of positively (R and K, including H when specifically indicated) or negatively (D and E) charged residues or net charge.

<sup>b</sup>Aromatic residues are W, Y, and F.

<sup>c</sup>High contents of charged residues, net charges, and aromatic residues are indicated by bold letters, with thresholds at 20%, 6%, and 8%, respectively, when measured as percentages of the sequence length.

<sup>d</sup>Phosphorylated Y was assigned a charge of -2 and assumed to be no longer an aromatic residue.

<sup>e</sup>Single-domain folded protein.

<sup>f</sup>H is treated as positively charged at the low pH condition.

**Table 1.**

**Correlation between class of salt dependence and amino-acid composition**

Two of the proteins included in **Table 1** are Caprin1 and its phosphorylated variant pY-Caprin1. Caprin1's phase separation required 500 mM NaCl and was promoted by higher salt,<sup>26</sup> and thus behaved as if in the high net charge class of salt dependence (**Figure 6A**, third panel). In contrast, pY-Caprin1's phase separation was suppressed by salt,<sup>18</sup> behaving as if in the screening class (**Figure 6A**, first panel). Our amino-acid composition-based predictor correctly places both proteins in these respective classes (**Table 1**), with Caprin1's sequence possessing both a high net charge and a high aromatic content whereas pY-Caprin1's sequence possessing a high level of charged residues. The disparate salt dependences of these two proteins were also correctly captured by both analytical theory and coarse-grained simulations.<sup>18</sup> The simulations also showed the dominance of  $\text{Cl}^-$  over  $\text{Na}^+$  in partitioning into the dense phase of Caprin1, similar to our all-atom simulations of A1-LCD (**Figure 3B**); both Caprin1 and A1-LCD have a high net positive charge.

Comparing and contrasting coarse-grained simulations and all-atom simulations highlight the pros and cons of the two approaches. Coarse-grained simulations can readily capture the phase-separation process and represent the equilibrium between the dense and dilute phases. For short peptides, all-atom simulations have been shown to accomplish both of these tasks.<sup>37</sup> For the mixture of ATP and a 33-residue IDP, all-atom simulations were able to capture the condensation process but not the equilibration of the resulting dense phase with a dilute phase.<sup>36</sup> In this work, although the total number of atoms in the A1-LCD system, ~half a million, is very high from a computational standpoint, the number of protein chains (i.e., 8) is still too low for a proper representation of a homogenous solution at low salt and a full equilibration between two phases at intermediate and high salt. Nevertheless, our simulations do not show condensation at low salt, thus mimicking a homogenous solution; the simulations at intermediate and high salt show condensation and thus model the dense phase. We are thus able to compare the salt effects in a homogenous solution (50 mM NaCl) and in the dense phase after phase separation (150-1000 mM NaCl) and compare the salt effects in dense phases with intermediate (150 and 300 mM NaCl) and high condensation (500 and 1000 mM NaCl). Importantly, the all-atom simulations are able to capture unusual physical effects that could be missed in coarse-grained simulations, such as strengthening  $\pi$ -types of interactions by drawing water away from the interaction partners. The latter effect is now confirmed by Raman spectroscopy.<sup>40</sup>

To conclude, salts regulate intermolecular interactions in biomolecular condensates in a variety of ways, but the net dependence of homotypic phase separation on salt concentration can be placed into four distinct classes (**Figure 6A**, B). Moreover, these classes are predictable from the protein amino-acid composition. Salts also exert significant effects on heterotypic condensates,<sup>8,19</sup> the conclusions drawn here on homotypic condensates may prove instructive for heterotypic condensates. The fact that different IDPs respond to salts differently raises interesting questions on phase separation inside cells. For example, a fluctuation in intracellular salt concentration may promote the phase separation of some IDPs and suppress the phase separation of other IDPs. If two IDPs co-phase separate or form heterotypic condensates, salt fluctuations may change the protein composition of condensates and the relative contributions of the individual proteins to phase separation. Lastly, we note that the atomistic details of ion coordination to A1-LCD backbone or sidechain groups are reminiscent of those observed in other systems, in particular channels and transporters for  $\text{Na}^+$  and  $\text{Cl}^-$ .<sup>42-44</sup>

## Computational Methods

### Molecular dynamics simulations

MD simulations were performed using AMBER 18<sup>45</sup> with ff14SB force field<sup>46</sup> for the protein and TIP4P-D for water.<sup>47</sup> The initial configuration of the 8-chain system was constructed using A1-LCD conformations from previous single-copy simulations.<sup>39</sup> Specifically, 4 copies with different conformations were placed in a rectangular box, positioned with minimal inter-chain

contacts within 3.5 Å. This 4-copy subsystem was duplicated to form the initial configuration of the 8-chain system (**Figure 1B**). The protein chains were solvated in a box with dimensions of 182 Å × 146 Å × 164 Å. For each desired salt concentration (50, 150, 300, 500, or 1000 mM), an appropriate number of water molecules were randomly selected and replaced with Cl<sup>-</sup> and Na<sup>+</sup> ions; excess Cl<sup>-</sup> was added to neutralize the system. The total number of atoms was ~500,000 for the 8-chain system at each salt concentration.

After energy minimization in sander (2000 steps of steepest descent and 3000 steps of conjugate gradient), each system was heated to 300 K over 100 ps with a 1 fs timestep, under constant NVT using the Langevin thermostat<sup>48</sup> with a 3.0 ps<sup>-1</sup> damping constant. The simulation was then continued in four replicates at constant NPT for 1.5 μs with a 2 fs timestep. The final dimensions were stabilized at ~174 Å × 140 Å × 157 Å. Pressure was regulated using the Berendsen barostat<sup>49</sup> with a coupling constant of 2.0 ps. All simulations were run on GPUs using *pmemd.cuda*<sup>50</sup>. Bond lengths involving hydrogens were constrained using the SHAKE algorithm.<sup>51</sup> Long-range electrostatic interactions were treated by the particle mesh Ewald method.<sup>52</sup> A cutoff distance of 10 Å was used for the nonbonded interactions. Frames were saved every 200 ps and the last 500 ns was used for analysis.

## Data analysis

$D_{\max}$  was calculated using the minmax function in VMD.<sup>53</sup> The differences between the maximum and minimum values of *x*, *y*, and *z* coordinates of all protein atoms were obtained; the largest of the three difference values was taken as  $D_{\max}$ . To avoid issues with periodic boundary conditions, this calculation was repeated 8 times, each time the simulation box was centered on one of the A1-LCD chains. The smallest of the 8  $D_{\max}$  values is reported in **Figure 2A**. Radius of gyration ( $R_g$ ) for each chain was calculated using the *radgyr* function in CPPTRAJ and then averaged over the 8 chains in each simulation (**Figure 2B**).<sup>54</sup> For inter- and intrachain contacts per residue (**Figure 2C**), the number of other residues in contact with a given residue was found using the distance mask function in CPPTRAJ with a cutoff of 6 Å between heavy atoms. Again, to avoid issues with the periodic boundary conditions, each chain was centered individually before inter- and intrachain contacts were calculated for each residue within that chain. In the case of intrachain contacts, the two nearest neighboring residues of a given residue were excluded as contact partners. Interchain contacts per chain (**Figure 5A**) were found in a similar way using only sidechain heavy atoms, and were broken into interaction types based on the residue types of the interaction partners. Interaction types were defined as follows: salt bridges were between Arg/Lys and Asp, cation- $\pi$  interactions were between Arg/Lys and Tyr/Phe, amino- $\pi$  interactions were between Asn/Gln and Tyr/Phe, and  $\pi$ - $\pi$  interactions were between a Tyr/Phe and another Tyr/Phe. To characterize chain reconfiguration, the RMSFs of the chains were calculated separately and then averaged over the 8 chains (Supplementary Figure 1).

Radial distribution functions (RDFs) for ions around protein atoms were calculated in CPPTRAJ using the *radial* command with a bin size of 0.05 Å and a specific selection of the protein atom type for either Cl<sup>-</sup> or Na<sup>+</sup> at the center. These RDF plots were used to select cutoffs for 1<sup>st</sup>- and 2<sup>nd</sup>-shell ion binding; the cutoffs were 4.0 and 6.4 Å, respectively, for Cl<sup>-</sup> and 3.0 and 5.4 Å, respectively, for Na<sup>+</sup> (Supplementary Figures 2 and 3). The number of 1<sup>st</sup>-shell or 1<sup>st</sup>- and 2<sup>nd</sup>-shell ions interacting with N and O atoms in a given sidechain type, the backbone, or the 8 chains together (**Figure 3**) was calculated using CPPTRAJ, by specifying a protein selection, an ion selection (all Cl<sup>-</sup> ions or all Na<sup>+</sup> ions), and a cutoff distance. Protein chains were centered one at a time and each time, the protein selection was limited to the central chain. The resulting ion-protein interactions for the 8 chains were then combined and custom python scripts were used to collect unique ions in the total ion count for each protein selection (**Figure 3A**, right ordinate; **Figure 3B**). For ions interacting with a given sidechain type, the total ion count was normalized by the number of residues of that type to get the ion count per residue (**Figure 3A**, left ordinate). The net charge of the system was calculated by taking the total charge of the 8 protein chains (+72) and adding and

subtracting, respectively, the numbers of bound  $\text{Na}^+$  and  $\text{Cl}^-$  ions within the 2<sup>nd</sup>-shell cutoffs (**Figure 3C** [↗](#)). The total bound ions in **Figure 3B** [↗](#) were further divided according to the number of chains with which each ion was interacting in the same frame. Ions were considered bridging if they interacted with two or more chains (based on 2<sup>nd</sup>-shell cutoffs) in the same frame (**Figure 4B** [↗](#)). Those interacting with an exact number of chains (e.g., 3) were identified when the same ion was found in the interaction lists of that many chains (**Figure 4C** [↗](#)). Error bars in all graphs represent the standard deviations among the four replicate simulations at each salt concentration.

RDFs for water molecules around Tyr residues were calculated using VMD with a bin size of 0.05 Å (**Figure 5E** [↗](#) and Supplementary Figure 4A-C). The water atom selection was all its three atoms and the Tyr atom selection was its 6-carbon ring. Moreover, only Tyr residues that interact with a specific residue type (within 6 Å of the 6-carbon ring) were selected: the partner residues were Phe (6-carbon ring; for  $\pi$ - $\pi$ ), Arg and Lys (sidechain N atoms; for cation- $\pi$ ), or Gln and Asn (sidechain N atoms; for amino- $\pi$ ). As a control, RDFs for water were also calculated around Asp sidechain oxygens that interact with Arg and Lys sidechain nitrogens (Supplementary Figure 4D).

## Acknowledgements

This work was supported by National Institutes of Health Grant GM118091.

## References

- (1) Mitrea D. M., Cika J. A., Stanley C. B., Nourse A., Onuchic P. L., Banerjee P. R., Phillips A. H., Park C.-G., Deniz A. A., Kriwacki R. W (2018) **Self-interaction of NPM1 modulates multiple mechanisms of liquid–liquid phase separation** *Nat Commun* **9**
- (2) Protter D. S. W., Parker R (2016) **Principles and Properties of Stress Granules** *Trends Cell Biol* **26**:668–679
- (3) Das S., Lin Y. H., Vernon R. M., Forman-Kay J. D., Chan H. S (2020) **Comparative Roles of Charge, Pi, and Hydrophobic Interactions in Sequence-Dependent Phase Separation of Intrinsically Disordered Proteins** *Proc Natl Acad Sci U S A* **117**:28795–28805
- (4) Dignon G. L., Best R. B., Mittal J (2020) **Biomolecular Phase Separation: From Molecular Driving Forces to Macroscopic Properties** *Annu Rev Phys Chem* **71**:53–75
- (5) Zhou H. X., Kota D., Qin S., Prasad R (2024) **Fundamental Aspects of Phase-Separated Biomolecular Condensates** *Chem Rev* **124**:8550–8595
- (6) Baldwin R. L (1996) **How Hofmeister ion interactions affect protein stability** *Biophys J* **71**:2056–2063
- (7) Zhou H. X., Pang X (2018) **Electrostatic Interactions in Protein Structure, Folding, Binding, and Condensation** *Chem Rev* **118**:1691–1741
- (8) Farag M., Borchers W. M., Bremer A., Mittag T., Pappu R. V (2023) **Phase separation of protein mixtures is driven by the interplay of homotypic and heterotypic interactions** *Nat Commun* **14**
- (9) Hazra M. K., Levy Y (2023) **Cross-Talk of Cation– $\pi$  Interactions with Electrostatic and Aromatic Interactions: A Salt-Dependent Trade-off in Biomolecular Condensates** *J Phys Chem Lett* **14**:8460–8469
- (10) Berry J., Weber S. C., Vaidya N., Haataja M., Brangwynne C. P (2015) **RNA transcription modulates phase transition-driven nuclear body assembly** *Proc Natl Acad Sci U S A* **112**:E5237–5245
- (11) Zhang H., Elbaum-Garfinkle S., Langdon E. M., Taylor N., Occhipinti P., Bridges A. A., Brangwynne C. P., Gladfelter A. S (2015) **RNA Controls PolyQ Protein Phase Transitions** *Mol Cell* **60**:220–230
- (12) Brady J. P., Farber P. J., Sekhar A., Lin Y. H., Huang R., Bah A., Nott T. J., Chan H. S., Baldwin A. J., Forman-Kay J. D. (2017) **Structural and hydrodynamic properties of an intrinsically disordered region of a germ cell-specific protein on phase separation** *Proc Natl Acad Sci U S A* **114**:E8194–E8203
- (13) Strom A. R., Emelyanov A. V., Mir M., Fyodorov D. V., Darzacq X., Karpen G. H (2017) **Phase separation drives heterochromatin domain formation** *Nature* **547**:241–245

- (14) Wei M. T., Elbaum-Garfinkle S., Holehouse A. S., Chen C. C., Feric M., Arnold C. B., Priestley R. D., Pappu R. V., Brangwynne C. P (2017) **Phase behaviour of disordered proteins underlying low density and high permeability of liquid organelles** *Nat Chem* **9**:1118–1125
- (15) Reed E. H., Hammer D. A (2018) **Redox sensitive protein droplets from recombinant oleosin** *Soft Matter* **14**:6506–6513
- (16) Tsang B., Arsenault J., Vernon R. M., Lin H., Sonenberg N., Wang L.-Y., Bah A., Forman-Kay J. D (2019) **Phosphoregulated FMRP Phase Separation Models Activity-Dependent Translation through Bidirectional Control of mRNA Granule Formation** *Proc. Natl. Acad. Sci. USA* **116**:4218–4227
- (17) Martin E. W., Thomasen F. E., Milkovic N. M., Cuneo M. J., Grace C. R., Nourse A., Lindorff-Larsen K., Mittag T (2021) **Interplay of folded domains and the disordered low-complexity domain in mediating hnRNP A1 phase separation** *Nucleic Acids Res* **49**:2931–2945
- (18) Lin Y.-H., Kim T. H., Das S., Pal T., Wessén J., Rangadurai A. K., Kay L. E., Forman-Kay J. D., Chan H. S (2024) **Electrostatics of Salt-Dependent Reentrant Phase Behaviors Highlights Diverse Roles of ATP in Biomolecular Condensates** *eLife* <https://doi.org/10.7554/elife.100284.1>
- (19) Galvanetto N., Ivanović M. T., Chowdhury A., Sottini A., Nüesch M. F., Nettels D., Best R. B., Schuler B (2023) **Extreme dynamics in a biomolecular condensate** *Nature* **619**:876–883
- (20) Muschol M., Rosenberger F (1997) **Liquid-liquid phase separation in supersaturated lysozyme solutions and associated precipitate formation/crystallization** *J Chem Phys* **107**:1953–1962
- (21) Burke K. A., Janke A. M., Rhine C. L., Fawzi N. L (2015) **Residue-by-Residue View of In Vitro FUS Granules that Bind the C-Terminal Domain of RNA Polymerase II** *Mol Cell* **60**:231–241
- (22) Kim S., Yoo H. Y., Huang J., Lee Y., Park S., Park Y., Jin S., Jung Y. M., Zeng H., Hwang D.S. (2017) **Salt Triggers the Simple Coacervation of an Underwater Adhesive When Cations Meet Aromatic pi Electrons in Seawater** *ACS Nano* **11**:6764–6772
- (23) Dao T. P., Kolaitis R. M., Kim H. J., O'Donovan K., Martyniak B., Colicino E., Hehnly H., Taylor J. P., Castaneda C. A (2018) **Ubiquitin Modulates Liquid-Liquid Phase Separation of UBQLN2 via Disruption of Multivalent Interactions** *Mol Cell* **69**:965–978
- (24) Babinchak W. M., Haider R., Dumm B. K., Sarkar P., Surewicz K., Choi J. K., Surewicz W. K (2019) **The role of liquid-liquid phase separation in aggregation of the TDP-43 low-complexity domain** *J Biol Chem* **294**:6306–6317
- (25) Le Ferrand H., Duchamp M., Gabryelczyk B., Cai H., Miserez A (2019) **Time-Resolved Observations of Liquid-Liquid Phase Separation at the Nanoscale Using in Situ Liquid Transmission Electron Microscopy** *J Am Chem Soc* **141**:7202–7210
- (26) Wong L. E., Kim T. H., Muhandiram D. R., Forman-Kay J. D., Kay L. E (2020) **NMR Experiments for Studies of Dilute and Condensed Protein Phases: Application to the Phase-Separating Protein CAPRIN1** *J Am Chem Soc* **142**:2471–2489
- (27) Agarwal A., Rai S. K., Avni A., Mukhopadhyay S (2021) **An intrinsically disordered pathological prion variant Y145Stop converts into self-seeding amyloids via liquid-liquid phase separation** *Proc Natl Acad Sci U S A* **118**

- (28) Krainer G., Welsh T. J., Joseph J. A., Espinosa J. R., Wittmann S., de Csillery E., Sridhar A., Toprakcioglu Z., Gudiskyte G., Czekalska M.A. (2021) **Reentrant liquid condensate phase of proteins is stabilized by hydrophobic and non-ionic interactions** *Nat Commun* **12**
- (29) Otis J. B., Sharpe S (2022) **Sequence Context and Complex Hofmeister Salt Interactions Dictate Phase Separation Propensity of Resilin-like Polypeptides** *Biomacromolecules* **23**:5225–5238
- (30) Martin E. W., Holehouse A. S., Peran I., Farag M., Incicco J. J., Bremer A., Grace C. R., Soranno A., Pappu R. V., Mittag T (2020) **Valence and patterning of aromatic residues determine the phase behavior of prion-like domains** *Science* **367**:694–699
- (31) Bremer A., Farag M., Borchers W. M., Peran I., Martin E. W., Pappu R. V., Mittag T (2022) **Deciphering how naturally occurring sequence features impact the phase behaviours of disordered prion-like domains** *Nat Chem* **14**:196–207
- (32) Tesei G., Lindorff-Larsen K (2023) **Improved Predictions of Phase Behaviour of Intrinsically Disordered Proteins by Tuning the Interaction Range** *Open Res Eur* **2**
- (33) Lin Y.-H., Forman-Kay J. D., Chan H. S (2016) **Sequence-Specific Polyampholyte Phase Separation in Membraneless Organelles** *Phys Rev Lett* **117**
- (34) Lin Y. H., Brady J. P., Chan H. S., Ghosh K (2020) **A unified analytical theory of heteropolymers for sequence-specific phase behaviors of polyelectrolytes and polyampholytes** *J Chem Phys* **152**
- (35) Garaizar A., Espinosa J. R (2021) **Salt dependent phase behavior of intrinsically disordered proteins from a coarse-grained model with explicit water and ions** *J Chem Phys* **155**
- (36) Kota D., Prasad R., Zhou H. X (2024) **Adenosine Triphosphate Mediates Phase Separation of Disordered Basic Proteins by Bridging Intermolecular Interaction Networks** *J Am Chem Soc* **146**:1326–1336
- (37) Zhang Y., Prasad R., Su S., Lee D., Zhou H.-X (2024) **Amino Acid-Dependent Phase Equilibrium and Material Properties of Tetrapeptide Condensates** *Cell Rep Phys Sci* **5**
- (38) Rekhi S., Garcia C. G., Barai M., Rizuan A., Schuster B. S., Kiick K. L., Mittal J (2024) **Expanding the molecular language of protein liquid-liquid phase separation** *Nat Chem* <https://doi.org/10.1038/s41557-024-01489-x>
- (39) Dey S., MacAinsh M., Zhou H. X (2022) **Sequence-Dependent Backbone Dynamics of Intrinsically Disordered Proteins** *J Chem Theory Comput* **18**:6310–6323
- (40) Joshi A., Avni A., Walimbe A., Rai S. K., Sarkar S., Mukhopadhyay S (2024) **Hydrogen-Bonded Network of Water in Phase-Separated Biomolecular Condensates** *J Phys Chem Lett* **15**:7724–7734
- (41) Orkoulas G., Kumar S. K., Panagiotopoulos A. Z (2003) **Monte carlo study of coulombic criticality in polyelectrolytes** *Phys Rev Lett* **90**
- (42) Mancusso R., Gregorio G. G., Liu Q., Wang D. N (2012) **Structure and mechanism of a bacterial sodium-dependent dicarboxylate transporter** *Nature* **491**:622–626

- (43) Mita K., Sumikama T., Iwamoto M., Matsuki Y., Shigemitsu K., Oiki S (2021) **Conductance selectivity of Na(+) across the K(+) channel via Na(+) trapped in a tortuous trajectory** *Proc Natl Acad Sci U S A* **118**
- (44) Leisle L., Lam K., Dehghani-Ghahnaviye S., Fortea E., Galpin J. D., Ahern C. A., Tajkhorshid E., Accardi A (2022) **Backbone amides are determinants of Cl(-) selectivity in CLC ion channels** *Nat Commun* **13**
- (45) Case D. A., Ben-Shalom I. Y., Brozell S. R., Cerutti D. S., Cheatham T. E., Cruzeiro V. W. D., Darden T. A., Duke R. E., Ghoreishi D., Gilson M. K. (2018) **Amber 2018 reference manual** San Francisco: University of California
- (46) Maier J. A., Martinez C., Kasavajhala K., Wickstrom L., Hauser K. E., Simmerling C (2015) **ff14SB: Improving the Accuracy of Protein Side Chain and Backbone Parameters from ff99SB** *J Chem Theory Comput* **11**:3696–3713
- (47) Piana S., Donchev A. G., Robustelli P., Shaw D. E (2015) **Water dispersion interactions strongly influence simulated structural properties of disordered protein states** *J Phys Chem B* **119**:5113–5123
- (48) Pastor R. W., Brooks B. R., Szabo A (1988) **An analysis of the accuracy of langevin and molecular dynamics algorithms** *Mol Phys* **65**:1409–1419
- (49) Berendsen H. J. C., Postma J. P. M., van Gunsteren W. F., DiNola A., Haak J. R (1984) **Molecular dynamics with coupling to an external bath** *J Chem Phys* **81**:3684–3690
- (50) Salomon-Ferrer R., Gotz A. W., Poole D., Le Grand S., Walker R. C (2013) **Routine Microsecond Molecular Dynamics Simulations with AMBER on GPUs. 2. Explicit Solvent Particle Mesh Ewald** *J Chem Theory Comput* **9**:3878–3888
- (51) Ryckaert J. P., Ciccotti G., Berendsen H. J. C (1977) **Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes** *J Comput Phys* **23**:327–341
- (52) Essmann U., Perera L., Berkowitz M. L., Darden T., Lee H., Pedersen L. G (1995) **A smooth particle mesh Ewald method** *J Chem Phys* **103**:8577–8593
- (53) Humphrey W., Dalke A., Schulten K (1996) **VMD: visual molecular dynamics** *J Mol Graph* **14**:33–38
- (54) Roe D. R., Cheatham T. E (2013) **3rd. PTRAJ and CPPTRAJ: Software for Processing and Analysis of Molecular Dynamics Trajectory Data** *J Chem Theory Comput* **9**:3084–3095

## Editors

Reviewing Editor

**Qiang Cui**

Boston University, Boston, United States of America

Senior Editor

**Qiang Cui**

Boston University, Boston, United States of America

## Reviewer #1 (Public review):

### Summary:

The authors examined the salt-dependent phase separation of the low-complexity domain of hnRN-PA1 (A1-LCD). Using all-atom molecular dynamics simulations, they identified four distinct classes of salt dependence in the phase separation of intrinsically disordered proteins (IDPs), which can be predicted based on their amino acid composition. However, the simulations and analysis, in their current form, are inadequate and incomplete.

### Strengths:

The authors attempt to unravel the mechanistic insights into the interplay between salt and protein phase separation, which is important given the complex behavior of salt effects on this process. Their effort to correlate the influence of salt on the low-complexity domain of hnRNPA1 (A1-LCD) with a range of other proteins known to undergo salt-dependent phase separation is an interesting and valuable topic.

### Weaknesses:

Based on the reviewer's assessment of the manuscript, the following points were raised:

- (1) The simulation duration is too short to draw comprehensive conclusions about phase separation.
- (2) There are concerns regarding the convergence of the simulations, particularly as highlighted in Figure 2A.
- (3) The simulation begins with a protein concentration of 3.5 mM ("we built an 8-copy model for the dense phase (with an initial concentration of 3.5 mM)"), which is high for phase separation studies. The reviewer questions the use of the term "dense phase" and suggests that the authors conduct a clearer analysis depicting the coexistence of both the dilute and dense phases to represent a steady state. Without this, the realism of the described phenomena is doubtful. Commenting on phase separation under conditions that don't align with typical phase separation parameters is not acceptable.
- (4) The inference that "Each Arg sidechain often coordinates two Cl<sup>-</sup> ions simultaneously, but each Lys sidechain coordinates only one Cl<sup>-</sup> ion" is questioned. According to Supplementary Figure 2A, Lys seems to coordinate with Cl<sup>-</sup> ions more frequently than Arg.
- (5) The authors are requested to update the figure captions for Supplementary Figures 2 and 3, specifying which system the analyses were performed on.
- (6) It is difficult to observe a clear trend due to irregularities in the data. Although the authors have included a red dotted line in the figures, the trend is not monotonic. The reviewer expresses concerns about significant conclusions drawn from these figures (e.g., Figure 2C, Figure 5A, Supplementary Figure 1).
- (7) Given the error in the radius of gyration (R<sub>g</sub>) calculations, the reviewer questions the validity of drawing conclusions from this data.
- (8) The pair correlation function values in Figure 5E and supplementary figure 4 show only minor differences, and the reviewer questions whether these differences are significant.
- (9) Previous reports suggest that, upon self-assembly, protein chains extend within the condensate, leading to a decrease in intramolecular contacts. However, the authors show an increase in intramolecular contacts with increasing salt concentration (Figure 2C), which contradicts prior studies. The reviewer advises the authors to carefully review this and provide justification.
- (10) A systematic comparison of estimated parameters with varying salt concentrations is required. Additionally, the authors should provide potential differences in salt concentrations between the dilute and condensed phases.
- (11) The reviewer finds that the majority of the data presented shows no significant alteration

with changes in salt concentration, yet the authors have made strong conclusions regarding salt activity.

The manuscript lacks sufficient scientific details of the calculations.

<https://doi.org/10.7554/eLife.100282.2.sa3>

**Reviewer #2 (Public review):**

This is an interesting computational study addressing how salt affects the assembly of biomolecular condensates. The simulation data are valuable as they provide a degree of atomistic details regarding how small salt ions modulate interactions among intrinsically disordered proteins with charged residues, namely via Debye-like screening that weakens the effective electrostatic interactions among the polymers, or through bridging interactions that allow interactions between like charges from different polymer chains to become effectively attractive (as illustrated, e.g., by the radial distribution functions in Supplementary Information). However, this manuscript has several shortcomings: (i) Connotations of the manuscript notwithstanding, many of the authors' concepts about salt effects on biomolecular condensates have been put forth by theoretical models, at least back in 2020 and even earlier. Those earlier works afford extensive information such as considerations of salt concentrations inside and outside the condensate (tie-lines). But the authors do not appear to be aware of this body of prior works and therefore missed the opportunity to build on these previous advances and put the present work with its complementary advantages in structural details in the proper context. (ii) There are significant experimental findings regarding salt effects on condensate formation [which have been modeled more recently] that predate the A1-LCD system (ref.19) addressed by the present manuscript. This information should be included, e.g., in Table 1, for sound scholarship and completeness. (iii) The strengths and limitations of the authors' approach vis-à-vis other theoretical approaches should be discussed with some degree of thoroughness (e.g., how the smallness of the authors' simulation system may affect the nature of the "phase transition" and the information that can be gathered regarding salt concentration inside vs. outside the "condensate" etc.).

Comments on revised version:

The authors have adequately addressed my previous concerns and suggestions. The manuscript is now significantly improved. The new results and analyses provided by the authors represent a substantial advance in our understanding of the role of electrostatics in the assembly of biomolecular condensates.

<https://doi.org/10.7554/eLife.100282.2.sa2>

**Reviewer #3 (Public review):**

Summary:

This study investigates the salt-dependent phase separation of A1-LCD, an intrinsically disordered region of hnRNPA1 implicated in neurodegenerative diseases. The authors employ all-atom molecular dynamics (MD) simulations to elucidate the molecular mechanisms by which salt influences A1-LCD phase separation. Contrary to typical intrinsically disordered protein (IDP) behavior, A1-LCD phase separation is enhanced by NaCl concentrations above 100 mM. The authors identify two direct effects of salt: neutralization of the protein's net charge and bridging between protein chains, both promoting condensation. They also uncover an indirect effect, where high salt concentrations strengthen pi-type interactions by

reducing water availability. These findings provide a detailed molecular picture of the complex interplay between electrostatic interactions, ion binding, and hydration in IDP phase separation.

**Strengths:**

- **Novel Insight:** The study challenges the prevailing view that salt generally suppresses IDP phase separation, highlighting A1-LCD's unique behavior.
- **Rigorous Methodology:** The authors utilize all-atom MD simulations, a powerful computational tool, to investigate the molecular details of salt-protein interactions.
- **Comprehensive Analysis:** The study systematically explores a wide range of salt concentrations, revealing a nuanced picture of salt effects on phase separation.
- **Clear Presentation:** The manuscript is well-written and logically structured, making the findings accessible to a broad audience.

**Weaknesses:**

- **Limited Scope:** The study focuses solely on the truncated A1-LCD, omitting simulations of the full-length protein. This limitation reduces the study's comparative value, as the authors note that the full-length protein exhibits typical salt-dependent behavior. However, given the much larger size of the full-length protein, it is acceptable to omit it given the current computing resources available.

Overall, this manuscript represents a significant contribution to the field of IDP phase separation. The authors' findings provide valuable insights into the molecular mechanisms by which salt modulates this process, with potential implications for understanding and treating neurodegenerative diseases.

<https://doi.org/10.7554/eLife.100282.2.sa1>

**Author response:**

The following is the authors' response to the original reviews.

**Reviewer #1 (Public Review):**

*Summary:*

*The authors examined the salt-dependent phase separation of the low-complexity domain of hnRN-PA1 (A1-LCD). Using all-atom molecular dynamics simulations, they identified four distinct classes of salt dependence in the phase separation of intrinsically disordered proteins (IDPs), which can be predicted based on their amino acid composition. However, the simulations and analysis, in their current form, are inadequate and incomplete.*

*Strengths:*

*The authors attempt to unravel the mechanistic insights into the interplay between salt and protein phase separation, which is important given the complex behavior of salt effects on this process. Their effort to correlate the influence of salt on the low-complexity domain of hnRNPA1 (A1-LCD) with a range of other proteins known to undergo salt-dependent phase separation is an interesting and valuable topic.*

*Weaknesses:*

*(1) The simulations performed are not sufficiently long (Figure 2A) to accurately comment on phase separation behavior. The simulations do not appear to have converged well,*

indicating that the system has not reached a steady state, rendering the analysis of the trajectories unreliable.

We have extended the simulations for an additional 500 ns, to 1500 ns. The last 500 ns show reasonably good convergence (see Figure 2A).

(2) The majority of the data presented shows no significant alteration with changes in salt concentration. However, the authors have based conclusions and made significant comments regarding salt activities. The absence of error bars in the data representation raises questions about its reliability. Additionally, the manuscript lacks sufficient scientific details of the calculations.

We have now included error bars. With the error bars, the salt dependences of all the calculated properties (exception for  $R_g$ ) show a clear trend. Additionally, we have expanded the descriptions of our calculations (p. 15-16).

(3) In Figures 2B and 2C, the changes in the radius of gyration and the number of contacts do not display significant variations with changes in salt concentration. The change in the radius of gyration with salt concentration is less than 1 Å, and the number of contacts does not change by at least 1. The authors' conclusions based on these minor changes seem unfounded.

The variation of  $\sim 1$  Å for the calculated  $R_g$  is similar to the counterpart for the experimental  $R_g$ . As for the number of contacts, note that this property is presented on a per-residue basis, so a value of 1 means that each residue picks up one additional contact, or each protein chain gains a total of 131 contacts, when the salt concentration is increased from 50 to 1000 mM.

#### **Reviewer #2 (Public Review):**

*This is an interesting computational study addressing how salt affects the assembly of biomolecular condensates. The simulation data are valuable as they provide a degree of atomistic details regarding how small salt ions modulate interactions among intrinsically disordered proteins with charged residues, namely via Debye-like screening that weakens the effective electrostatic interactions among the polymers, or through bridging interactions that allow interactions between like charges from different polymer chains to become effectively attractive (as illustrated, e.g., by the radial distribution functions in Supplementary Information). However, this manuscript has several shortcomings:*

*(i) Connotations of the manuscript notwithstanding, many of the authors' concepts about salt effects on biomolecular condensates have been put forth by theoretical models, at least back in 2020 and even earlier. Those earlier works afford extensive information such as considerations of salt concentrations inside and outside the condensate (tie-lines). But the authors do not appear to be aware of this body of prior works and therefore missed the opportunity to build on these previous advances and put the present work with its complementary advantages in structural details in the proper context.*

*(ii) There are significant experimental findings regarding salt effects on condensate formation [which have been modeled more recently] that predate the A1-LCD system (ref.19) addressed by the present manuscript. This information should be included, e.g., in Table 1, for sound scholarship and completeness.*

*(iii) The strengths and limitations of the authors' approach vis-à-vis other theoretical approaches should be discussed with some degree of thoroughness (e.g., how the smallness of the authors' simulation system may affect the nature of the "phase*

transition" and the information that can be gathered regarding salt concentration inside vs. outside the "condensate" etc.). Accordingly, this manuscript should be revised to address the following. In particular, the discussion in the manuscript should be significantly expanded by including references mentioned below as well as other references pertinent to the issues raised.

(1) The ability to use atomistic models to address the questions at hand is a strength of the present work. However, presumably because of the computational cost of such models, the "phase-separated" "condensates" in this manuscript are extremely small (only 8 chains). An inspection of Fig.1 indicates that while the high-salt configuration (snapshot, bottom right) is more compact and droplet-like than the low-salt configuration (top right), it is not clear that the 50 mM NaCl configuration can reasonably correspond to a dilute or homogeneous phase (without phase separation) or just a condensate with a lower protein concentration because the chains are still highly associated. One may argue that they become two droplets touching each other (the chains are not fully dispersed throughout the simulation box, unlike in typical coarse-grained simulations of biomolecular phase separation). While it may not be unfair to argue from this observation that the condensed phase is less stable at low salt, this raises critical questions about the adequacy of the approach as a stand-alone source of theoretical information. Accordingly, an informative discussion of the limitation of the authors' approach and comparisons with results from complementary approaches such as analytical theories and coarsegrained molecular dynamics will be instructive-even imperative, especially since such results exist in the literature (please see below).

We now discuss the limitations of our all-atom simulations and also other approaches (p. 13; see below).

(2) The aforementioned limitation is reflected by the authors' choice of using  $D_{max}$  as a sort of phase separation order parameter. However, no evidence was shown to indicate that  $D_{max}$  exhibits a twostate-like distribution expected of phase separation. It is also not clear whether a  $D_{max}$  value corresponding to the linear dimension of the simulation box was ever encountered in the authors' simulated trajectories such that the chains can be reliably considered to be essentially fully dispersed as would be expected for the dilute phase. Moreover, as the authors have noted in the second paragraph of the Results, the variation of  $D_{max}$  with simulation time does not show a monotonic rank order with salt concentration. The authors' explanation is equivalent to stipulating that the simulation system has not fully equilibrated, inevitably casting doubt on at least some of the conclusions drawn from the simulation data.

First off, with the extended simulations, the  $D_{max}$  values converge to a tiered order rank, with successively decreasing values from low salt (50 mM) to intermediate salt (150 and 300 mM) to high salt (500 and 1000 mM). Secondly, as we now state (p. 13), our low-salt simulations mimic a homogenous solution whereas our high-salt simulations mimic the dense phase of a phase-separated system. The intermediate-salt simulations also mimic the dense phase but at a somewhat lower concentration (hence the intermediate  $D_{max}$  value).

(3) With these limitations, is it realistic to estimate possible differences in salt concentration between the dilute and condensed phases in the present work? These features, including tie-lines, were shown to be amenable to analytical theory and coarse-grained molecular dynamics simulation (please see below).

The differences in salt effects that we report do not represent those between two phases. Rather, as explained in the preceding reply, they represent differences between a

homogenous solution at low salt and the dense phase at higher salt. We also acknowledge salt effects calculated by analytical theory and coarse-grained simulations (p. 13).

*(4) In the comparison in Fig.2B between experimental and simulated radius of gyration as a function of [NaCl], there is an outlier among the simulated radii of gyration at [NaCl] ~ 250 mM. An explanation should be offered.*

After extending the simulations and analyzing the last 500 ns, the Rg data no longer show an outlier though still have some fluctuations from one salt concentration to another.

*(5) The phenomenon of no phase separation at zero and low salt and phase separation at higher salt has been observed for the IDP Caprin1 and several of its mutants [Wong et al., J Am Chem Soc 142, 24712489 (2020) [<https://pubs.acs.org/doi/full/10.1021/jacs.9b12208>], see especially Fig.9 of this reference]. This work should be included in the discussion and added to Table 1.*

We now have added Caprin1 to Table 1 (new ref 26) and discuss this paper (p. 13).

*(6) The authors stated in the Introduction that "A unifying understanding of how salt affects the phase separation of IDPs is still lacking". While it is definitely true that much remains to be learned about salt effects on IDP phase separation, the advances that have already been made regarding salt effects on IDP phase separation is more abundant than that conveyed by this narrative. For instance, an analytical theory termed rG-RPA was put forth in 2020 to provide a uniform (unified) treatment of salt, pH, and sequence-charge-pattern effects on polyampholytes and polyelectrolytes (corresponding to the authors' low net charge and high net charge cases). This theory offers a means to predict salt-IDP tie-lines and a comprehensive account of salt effect on polyelectrolytes resulting in a lack of phase separation at extremely low salt and subsequent salt-enhanced phase separation (similar to the case the authors studied here) and in some cases re-entrant phase separation or dissolution [Lin et al., J Chem Phys 152. 045102 (2020) [<https://doi.org/10.1063/1.5139661>]]. This work is highly relevant and it already provided a conceptual framework for the authors' atomistic results and subsequent discussion. As such, it should definitely be a part of the authors' discussion.*

We now cite this paper (new ref 34) in Introduction (p. 4). We also discuss its results for Caprin1 (new ref 18; p. 13).

*(7) Bridging interactions by small ions resulting in effective attractive interactions among polyelectrolytes leading to their phase separation have been demonstrated computationally by Orkoulas et al., Phys Rev Lett 90, 048303 (2003) [<https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.90.048303>]. This result should also be included in the discussion.*

We now cite this paper (new ref 41; p. 11).

*(8) More recently, the salt-dependent phase separations of Caprin1, its RtoK variants and phosphorylated variant (see item #5 above) were modeled (and rationalized) quite comprehensively using rG-RPA, field-theoretic simulation, and coarse-grained molecular dynamics [Lin et al., arXiv:2401.04873 [<https://arxiv.org/abs/2401.04873>]], providing additional data supporting a conceptual perspective put forth in Lin et al. J Chem Phys 2020 (e.g., salt-IDP tie-lines, bridging interactions, reentrance behaviors etc.) as well as in the authors' current manuscript. It will be very helpful to the readers of eLife to include this preprint in the authors' discussion, perhaps as per the authors' discretion along the*

manner in which other preprints are referenced and discussed in the current version of the manuscript.

We now cite this paper (new ref 18) and discuss it along with new ref 26 in Discussion (p. 13).

**Reviewer #3 (Public Review):**

*Summary:*

*This study investigates the salt-dependent phase separation of A1-LCD, an intrinsically disordered region of hnRNPA1 implicated in neurodegenerative diseases. The authors employ all-atom molecular dynamics (MD) simulations to elucidate the molecular mechanisms by which salt influences A1-LCD phase separation. Contrary to typical intrinsically disordered protein (IDP) behavior, A1-LCD phase separation is enhanced by NaCl concentrations above 100 mM. The authors identify two direct effects of salt: neutralization of the protein's net charge and bridging between protein chains, both promoting condensation. They also uncover an indirect effect, where high salt concentrations strengthen pi-type interactions by reducing water availability. These findings provide a detailed molecular picture of the complex interplay between electrostatic interactions, ion binding, and hydration in IDP phase separation.*

*Strengths:*

*Novel Insight: The study challenges the prevailing view that salt generally suppresses IDP phase separation, highlighting A1-LCD's unique behavior.*

*Rigorous Methodology: The authors utilize all-atom MD simulations, a powerful computational tool, to investigate the molecular details of salt-protein interactions.*

*Comprehensive Analysis: The study systematically explores a wide range of salt concentrations, revealing a nuanced picture of salt effects on phase separation.*

*Clear Presentation: The manuscript is well-written and logically structured, making the findings accessible to a broad audience.*

*Weaknesses:*

*Limited Scope: The study focuses solely on the truncated A1-LCD, omitting simulations of the full-length protein. This limitation reduces the study's comparative value, as the authors note that the full-length protein exhibits typical salt-dependent behavior. A comparative analysis would strengthen the manuscript's conclusions and broaden its impact.*

Perhaps we did not impress on the reviewer how expensive the all-atom MD simulations on A1-LCD were: the systems each contained half a million atoms and the simulations took many months to complete. That said, we agree with the reviewer that, ideally, a comparative study on a protein showing the typical screening class of salt dependence would have made our work more complete. However, we are confident of the conclusions for several reasons. First, the three salt effects – charge neutralization, bridging, and strengthening of pi-types of interactions – revealed by the all-atom simulations are physically sound and well-supported by other studies. Second, these effects led us to develop a unified picture for the salt dependence of homotypic phase separation, in the form of a predictor for the classes of salt dependence based on amino-acid composition. This predictor works well for nearly 30 proteins. Third, recent studies using analytical theory and coarse-grained simulations (new ref 18) also strongly support our conclusions.

**Reviewer #1 (Recommendations For The Authors):**

*(1) In Figure 1, the color scheme should be updated and the figure remade, as the current set of color choices makes it very difficult to distinguish the magenta spheres.*

We have increased the sizes of ions in Figure 1 to make them distinguishable.

*(2) Within the framework of atomistic simulations, the influence of salt concentration alteration on protein conformational plasticity is worth investigating. This could be correlated (with proper details) with the effect of salt-concentration-modulated protein aggregation behavior.*

We now use RMSF to measure conformational plasticity, which shows a clear salt-dependent trend with a 27% reduction in fluctuations from 50 mM to 1000 mM NaCl (new Fig. S1).

*(3) The authors should mention the protein concentrations employed in the simulations and whether these are consistent with experimentally used concentrations.*

We have mentioned the initial concentration (3.5 mM). We now further state that this concentration is maintained in the low-salt simulations, indicating absence of phase separation, but is increased to 23 mM in the high-salt simulations, indicating phase separation. The latter value is consistent with the measured concentrations in the dense phase (last two paragraphs of p. 5).

*(4) It would be useful to test the salt effect for at least two extreme salt concentrations at various protein concentrations, consistent with experimental protein concentration ranges.*

In simulation studies of short peptides (ref 37), we have shown that the initial concentration does not affect the final concentration in the dense phase, as expected for phase-separation systems. We expect that the same will be true for the A1-LCD system at intermediate and high salt where phase separation occurs. Though this expectation could be tested by simulations at a different initial protein concentration, such simulations would be expensive but unlikely to yield new physical insight.

*(5) Importantly, the simulations do not appear to have converged well enough (Figure 2A). The authors should extend the simulation trajectories to ensure the system has reached a steady state.*

We extended the simulations for an additional 500 ns, which now appear to show convergence. In Figure 2A we now see Dmax values converge to a tiered order rank, with successively decreasing values from low salt (50 mM) to intermediate salt (150 and 300 mM) to high salt (500 and 1000 mM).

*(6) The authors mention "phase separation" in the title, but with only a 1  $\mu$ s simulation trajectory, it is not possible to simulate a phenomenon like phase separation accurately. Since atomistic simulations cannot realistically capture phase separation on this timescale, a coarse-grained approach is more suitable. To properly explore salt effects in the context of phase separation, long timescale simulation trajectories should be considered. Otherwise, the data remain unreliable.*

Our all-atom simulations revealed rich salt effects that might have been missed in coarse-grained simulations. It is true that coarse-grained models allow the simulations of the phase separation process, but as we have recently demonstrated (refs 36 and 37), all-atom

simulations on the  $\mu$ s timescale are also able to capture the spontaneous phase separation of peptides and small IDPs. A1-LCD is much larger than those systems, so we had to use a relatively small chain number (8 chains here vs 64 used in ref 37 and 16 used in ref 37). S21l, we observe the condensation into a dense phase at high salt. We discuss the pros and cons of all-atom vs. coarse-grained simulations in p. 13.

*(7) In Figure 5E, the plot does not show that  $g(r)$  has reached 1. If it does, the authors should show the full curve. The same issue remains with supplementary figures 1, 2, 3, etc.*

We now show the approach to 1 in the insets of Figs. S2, S3, S4, and 5E.

*(8) None of the data is represented with error bars. The authors should include error bars in their data representations.*

We have now included error bars in all graphs that report average values.

*(9) The authors state that "the net charge of the system reduces to only +8 at 1000 mM NaCl (Figure 3C)" but do not explain how this was calculated.*

We now add this explanation in methods (p. 16).

*(10). The authors mention "similar to the role played by ATP molecules in driving phase separation of positively charged IDPs." However, ATP can inhibit aggregation, and its induction of phase separation is concentration-dependent. Given ATP's large aromatic moiety, its comparison to ions is not straightforward and is more complex. This comparison can be at best avoided.*

In this context we are comparing the bridging capability of ATP molecules in driving phase separation of positively charged IDPs in ref 36 to the bridging capability of the ions here. In ref 36 the authors show ATP bridging interactions between protein chains similar to what we show here with ions.

*(11) Many calculations are vaguely represented. The process for calculating the number of bridging ions, for example, is not well documented. The authors should provide sufficient details to allow for the reproducibility of the data.*

We have now expanded the methods section to include more detailed information on calculations done.

#### **Reviewer #3 (Recommendations For The Authors):**

*Include error bars or standard deviations for all results averaged over four replicates, particularly for the number of ions and contacts per residue. This would provide a clearer picture of the data's reliability and variability.*

We have now included error bars in all graphs that report averaged values.

*Strengthen the support for the conclusion that "each Arg sidechain often coordinates two  $\text{Cl}^-$  ions, multiple backbone carbonyls often coordinate a single  $\text{Na}^+$  ion." While Fig. 3A clearly demonstrates Arg $\text{Cl}^-$  coordination, the  $\text{Na}^+$  coordination claim for a 131-residue protein requires further clarification. Consider including the integration profile of radial distribution functions for  $\text{Na}^+$  ions to bolster this assertion.*

We now report the number of Na<sup>+</sup> ions that coordinate with multiple backbone carbonyls (p. 7) as well as the number of Na<sup>+</sup> ions that bridge between A1-LCD chains via coordination with multiple backbone carbonyls (p. 9). Please note that Figure 4A right panel displays an example of Na<sup>+</sup> coordinating with multiple backbone carbonyls.

*Address the following typographical errors in the main text: o Page 11, line 25: "distinct classes of sat dependence" should be "distinct classes of salt dependence" o Page 14, line 9: "for Cl- and 3.0 and 5.4 Å" should be "for Cl- and 3.0 and 5.4 Å" o Page 14, line 18: "As a control, PRDFs for water were also calculated" should be "As a control, RDFs for water were also calculated" (assuming PRDF was meant to be RDF)*

We have now corrected these typos.

*Consider expanding the study to include simulations of the full-length protein to provide a more comprehensive comparison between the truncated A1-LCD and the complete protein's behavior in various salt concentrations.*

As we explained above, even with eight chains of A1-LCD, which has 131 residues, the systems already contain half a million atoms each and the all-atom simulations took many months to complete. Full-length A1 has 314 residues so a multi-chain system would be too large to be feasible for all-atom simulations.

<https://doi.org/10.7554/eLife.100282.2.sa0>